NEAR REAL-TIME DNS THREAT DETECTION & CONTEXTUAL THREAT INTELLIGENCE

R-25-26J-066

PROJECT SUPERVISORS

SUPERVISOR

Mr. Kavinga Yapa

CO-SUPERVISOR

Ms. Ayesha Wijesooriya

RESEARCH GAP & OBJECTIVE

Problem: Existing DNS security solutions lack real-time detection, contextual intelligence, and efficient handling of large DNS log volumes required for ML-based analysis.

Main Objective: Develop an ML-based near real-time DNS anomaly detection system with centralized monitoring and contextual threat intelligence for enterprise environments.

SOLUTION

Solution Summary

AI/ML-powered platform for near real-time DNS threat detection with centralized monitoring and contextual threat intelligence.

Key features delivered

- Real-time DNS anomaly detection
- SIEM/SOC integrated monitoring
- Customizable detection with human intelligence
- Contextual threat intelligence

Key features delivered

- **High-performance:** near real-time processing
- **Scalable** log ingestion and analysis pipeline
- **Reliable** detection with explainability support
- **Analyst-friendly** investigation interface
- **Secure by Design:** ensuring CIA with AAA

PP2 COMMENTS

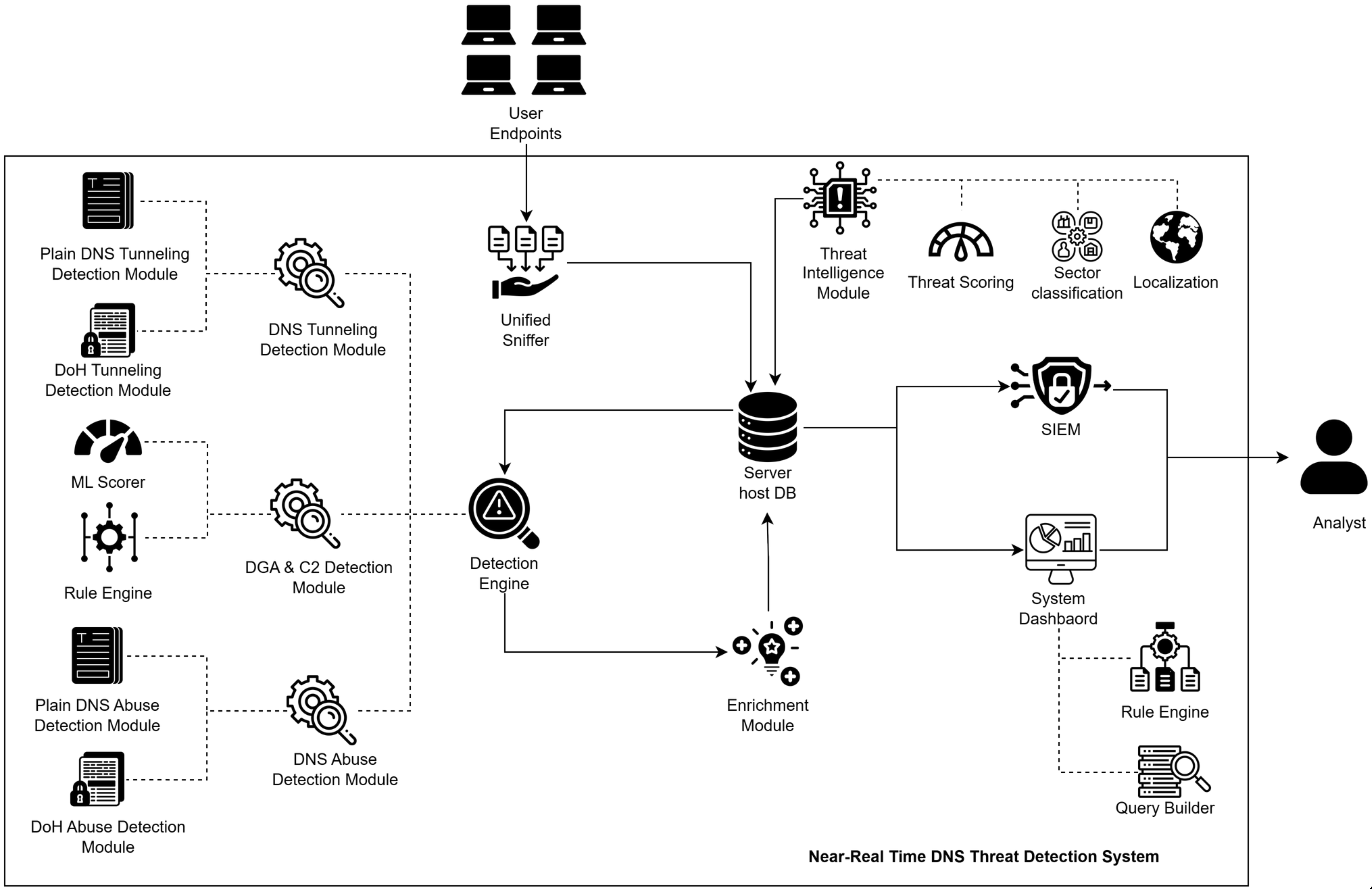
CONDUCT & SHOWCASE A FULL COMPARISON WITH EXISTING RESEARCH

CONDUCT A VAPT ON THE DASHBOARD AS WAS MENTIONED BY THE TEAM

OVERALL DESIGN EXCELLENCE

- **All-in-one DNS visibility:** covers tunneling, DGA/C2, DNS abuse, and encrypted DNS behavior.
- **Near-real-time detection flow:** captures, analyzes, and generates alerts continuously.
- **Hybrid detection design:** combines ML models, behavioral rules, and flow-based analysis.
- **Threat intelligence enrichment:** adds OSINT, IoCs, CVE context, IP reputation, and abuse history.
- **Investigation-ready events:** converts raw DNS logs into risk-scored host-level events.
- **C2 infrastructure mapping:** visualizes suspicious domain–IP relationships.
- **SOC-ready output:** supports dashboards and SIEM-friendly structured logs.

FULL SYSTEM ARCHITECTURE



AI/ML DETECTION OF DNS TUNNELING & DATA EXFILTRATION

COMPONENT FOCUS

- Detects covert data exfiltration hidden inside DNS traffic
- Supports both plaintext DNS and encrypted DNS-over-HTTPS traffic
- Converts raw query/flow detections into meaningful security events

NOVELTY / CONTRIBUTION

- Combines lexical, behavioural, and flow-based analysis for broader tunneling detection
- Uses model fusion for plaintext DNS to balance domain structure and traffic behaviour
- Detects DoH tunneling without payload inspection using flow-level statistical features
- Reduces alert noise through event aggregation, risk scoring, and SIEM-ready outputs

AI/ML DETECTION OF MALWARE C2 & DGA-BASED DNS

COMPONENT FOCUS

- Detects malware communication hidden within DNS traffic
- Identifies DGA-generated domains and DNS-based C2 behaviour
- Converts raw DNS activity into meaningful host-level security events

NOVELTY / CONTRIBUTION

- Combines domain-level ML detection with host-level behavioural correlation
- Uses suspicious domain scores to guide C2 rule evaluation
- Reduces alert noise through event aggregation instead of isolated query alerts
- Adds investigation context through threat intelligence and C2 infrastructure mapping

AI/ML DETECTION OF DNS ABUSE & INFRASTRUCTURE ATTACKS

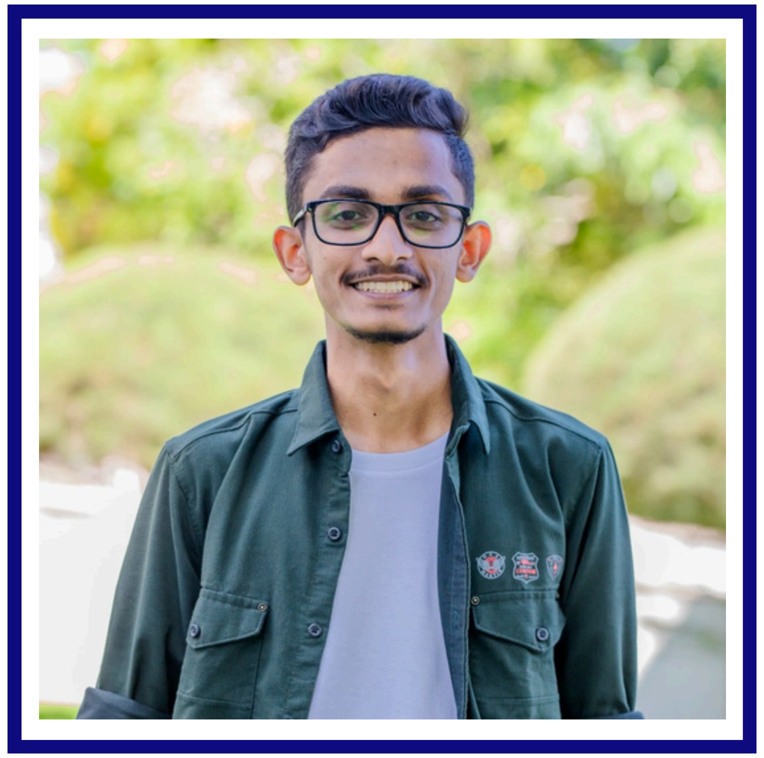
COMPONENT FOCUS

- Detects DNS abuse attacks across unencrypted DNS, DoH, and DoT traffic
- Identifies flooding, amplification/reflection, spoofing, and encrypted DNS abuse patterns
- Converts flow-level traffic into SOC-ready attack alerts and rolling verdicts

NOVELTY / CONTRIBUTION

- Uses a lightweight endpoint DPI agent to extract DNS-specific flow features
- Detects encrypted DNS abuse without payload decryption using TLS/flow metadata
- Applies XGBoost with F-beta threshold tuning to prioritise attack detection
- Supports live endpoint deployment with MySQL logging and Kibana/SOC integration

THREAT INTELLIGENCE & RESILIENCE MODULE



COMPONENT FOCUS

- Converts unstructured Surface Web and Dark Web threat reports into structured CTI
- Extracts targeted location, industry sector, IoCs, and vulnerability context
- Produces prioritized intelligence for SOC investigation and SIEM integration

NOVELTY / CONTRIBUTION

- Uses fine-tuned DistilBERT models to classify geopolitical targets and industry sectors
- Combines AI-derived context with NVD vulnerability scores and VirusTotal validation
- Introduces a confidence-weighted risk scoring engine with explainable score drivers
- Uses a distributed Tor “Scout Node” to safely collect Dark Web intelligence

COMMERCIALIZATION

ENTERPRISE

MSSPS / ISPS

MULTI-TENANT MONITORING, CUSTOM RULES, ADVANCED REPORTING, API INTEGRATION

PROFESSIONAL

SOC TEAMS / ENTERPRISES

ML DETECTION, EVENT AGGREGATION, SIEM EXPORT, THREAT INTELLIGENCE

STARTER

SMALL ENTERPRISES / UNIVERSITIES

DNS MONITORING, BASIC ALERTS, DASHBOARD

REVENUE MODEL

- Annual subscription based on number of endpoints or DNS query volume
- Integration fee for SIEM/SOC onboarding
- Custom rule development for enterprise customers
- Managed monitoring option for organizations without a dedicated SOC

Up to 10 endpoints

Starter

$299/year

Up to 100 endpoints

Professional

$999/year

Custom no of endpoints

Enterprise

Custom pricing

RESULTS

Detection Area	Deployment Support	Accuracy (%)	Precision (%)	Recall (%)	F1Score (%)
DNS Tunneling - Plain DNS	Real-time, DoH, SIEM	92.9	92.8	**93**	92.9
DNS Tunneling - DoH	Real-time, DoH, SIEM	91.4	91	**99.7**	94.9
DGA and C2 Detection	Real-time, SIEM	84.46	86.97	**88.9**	84.2
DNS Abuse Detection	Real-time, DoH/DoT, SIEM	94.5	98.72	**93.48**	96.03
TI - Country Classification	SIEM Ready	81.12	81	**81**	80
TI - Sector Classification	SIEM Ready	80.13	80	**80**	80

RESEARCH COMPARISON

DNS TUNNELING AND DATA EXFILTRATION DETECTION

Study / Reference	Methodology	Data Source	Environment	Real-Time Capability	Encrypted DNS Support	SIEM Integration	Accuracy	Precision	Recall	F1-Score
Chen et al. (2021)	LSTM (lexical / sequence-based)	Mixed traffic + tunneling tools	Lab-based testing	Partial (lab only)	Not supported	Not supported	0.993	0.991	0.996	0.993
Liu et al. (2017)	Behavioural ML (SVM, LR, DT)	Real DNS + injected tunnels	Recursive DNS deployment	Near real-time	Not supported	Not supported	0.999	0.9998	0.9993	0.9995
Altuncu et al. (2021)	Deep Neural Network (DNN)	Academic network + injected tunnels	Live network (controlled)	Strong real-time	Not supported	Not supported	0.999	0.998	0.999	0.9985
Zhan et al. (2022)	Flow-based ML (Random Forest)	Real DoH traffic	Partial deployment	Partial (DoH only)	DoH only	Not supported	–	–	~0.990	~0.990
Liang et al. (2023)	CNN + Clustering	Mixed datasets + real traffic	Partial deployment	Partial	Not supported	Not supported	–	~0.992	~0.992	~0.992
Proposed System – Plain DNS	**DST Fusion (Lexical + Behavioural LR)**	**Live captured DNS traffic**	**Real-time deployment**	**Fully supported**	**Plain DNS only**	**Supported (structured alerts)**	**0.929**	**0.928**	**0.93**	**0.929**
Proposed System – DoH	**Random Forest (flow-based features)**	**Live captured DoH traffic**	**Real-time deployment**	**Fully supported**	**Supported (DoH)**	**Supported (structured alerts)**	**0.914**	**0.91**	**0.997**	**0.949**

DGA DETECTION AND C2 APPROACHES

Study / Reference	Methodology	Data Source	Environment	Real-Time Capability	SIEM Integration	Accuracy	Precision	Recall	F1-Score
Li et al. (2019)	RNN + N-gram + Ensemble	Public DGA datasets (static benchmark)	Offline evaluation	Not supported	Not supported	0.9982	0.9982	0.9982	0.9982
Namgung et al. (2021)	CNN + BiLSTM Ensemble	Public DGA datasets (static benchmark)	Offline evaluation	Not supported	Not supported	0.9684	0.9676	0.9682	0.9666
Highnam et al. (2021)	CNN + LSTM Hybrid	Dictionary DGA + Alexa domains	Enterprise network validation	Not supported	Not supported	0.9656	0.9557	0.9766	0.966
Cruciani et al. (2021)	2-gram MLP Autoencoder	Legitimate domains + DGA dataset	Offline evaluation	Not supported	Not supported	N/A	0.67	0.8	0.73
Proposed System	**TF-IDF + Logistic Regression**	**Live captured DNS traffic (endpoint-based)**	**Real-time deployment**	**Supported (continuous capture)**	**Supported (structured logs)**	**0.8446**	**0.8697**	**0.8899**	**0.842**

DNS ABUSE DETECTION APPROACHES

Study / Reference	Methodology	Data Source	Environment	Real-Time Capability	Encrypted DNS Support	SIEM Integration	Accuracy	Precision	Recall	F1-Score
Sommese et al. (2020)	Traffic correlation (Backscatter + active probes)	UCSD Darknet (17-month) + OpenINTEL	Offline retrospective	No	No	No	N/A (No ML)	N/A	N/A	N/A
Suvra et al. (2021)	Traditional ML (RF, DT, AdaBoost)	CIC-DDoS2019	Simulated inference	No	No	No	0.988	0.99	0.98	0.98
Erskine (2023)	Deep Learning + MLP (DLMIDPSM)	CICIoT2023	Near real-time IoT IDPS	Yes	No	No	0.85	0.81	0.83	0.8
El Attar et al. (2023)	Supervised ML + Wrapper/Filter feature selection	CIC-DDoS2019 & 100k Real DNS Packets	Near real-time network layer	Yes	No	No	N/A	N/A	N/A	1.00 (XGB)
Alharby (2025)	ML + Undersampling + PCA/Chi-Square	CICIoT2023 (Undersampled)	Static high-speed	Yes	No	No	0.9999	0.9999	0.9999	0.9999
Proposed System	**XGBoost (39-feature DPI) + F-beta(3.0) tuning**	**Live Endpoint Capture**	**True Real-Time (Agent + SOC)**	**Yes**	**Yes (DoH/DoT)**	**Yes**	**0.9405**	**0.9872**	**0.9348**	**0.9603**

CONTEXTUAL THREAT INTELLIGENCE APPROACHES

Study / Approach	Methodology	Data Source	SIEM Integration	Accuracy	Precision	Recall	F1-Score
P. Balasubramanian et al. TSTEM (Sentence	NLP-based classification	Curated CTI reports	Not supported	0.98	0.98	0.96	0.97
P. Balasubramanian et al. TSTEM (Webpage	NLP-based classification	Curated CTI reports	Not supported	0.95	0.96	0.92	0.94
P. Balasubramanian et al. TSTEM (IOC Extraction - NER)	Named Entity Recognition	Curated CTI reports	Not supported	0.98	0.83	0.85	0.84
Proposed System - Sector Classification	**NLP-based classification**	**Live CTI feeds + reports**	**Supported (sector tagging)**	**0.8**	**0.8**	**0.8**	**0.8**
Proposed System - Location Classification	**NLP-based classification**	**Live CTI feeds + reports**	**Supported (geo tagging)**	**0.81**	**0.81**	**0.81**	**0.8**

LATENCY COMPARISON - CTI

DARKWEB

	Scraping Latency	Network Latency	AI Processing	VT & DB Save	Total Latency
	1.02	0.43	1.65	18.75	21.85
	1.93	0.40	0.89	1.76	4.98
	1.31	0.45	1.18	2.48	5.42
	1.71	0.44	1.31	2.23	5.69
	1.42	0.42	1.41	2.24	5.49
	1.01	0.43	1.11	2.08	4.63
	1.28	1.21	1.85	2.36	6.70
	1.21	1.20	2.00	2.52	6.93
	1.01	1.21	1.71	2.23	6.16
	1.35	1.18	1.85	2.39	6.77
	1.51	1.21	1.81	2.34	6.87
	1.66	1.21	1.83	2.37	7.07
Avg(Secs)	**1.37**	**0.82**	**1.55**	**3.65**	**7.38**

SURFACEWEB

	Scraping Latency	Network Latency	AI Processing	VT & DB Save	Total Latency
	0.90	0.00	2.06	19.12	22.08
	1.01	0.00	2.45	3.35	6.81
	0.99	0.00	0.94	1.83	3.76
	0.97	0.00	0.92	17.14	19.03
	1.04	0.00	2.59	65.32	68.95
	0.93	0.00	2.22	7.79	10.94
	1.41	0.00	9.28	16.56	27.25
	1.20	0.00	1.78	0.53	3.51
	1.17	0.00	1.34	0.56	3.07
	1.19	0.00	3.38	16.62	21.19
	1.04	0.00	5.03	0.62	6.69
	1.00	0.00	1.94	35.39	38.33
Avg(Secs)	**1.07**	**0.00**	**2.83**	**15.40**	**19.30**

THANK YOU!

DEMO

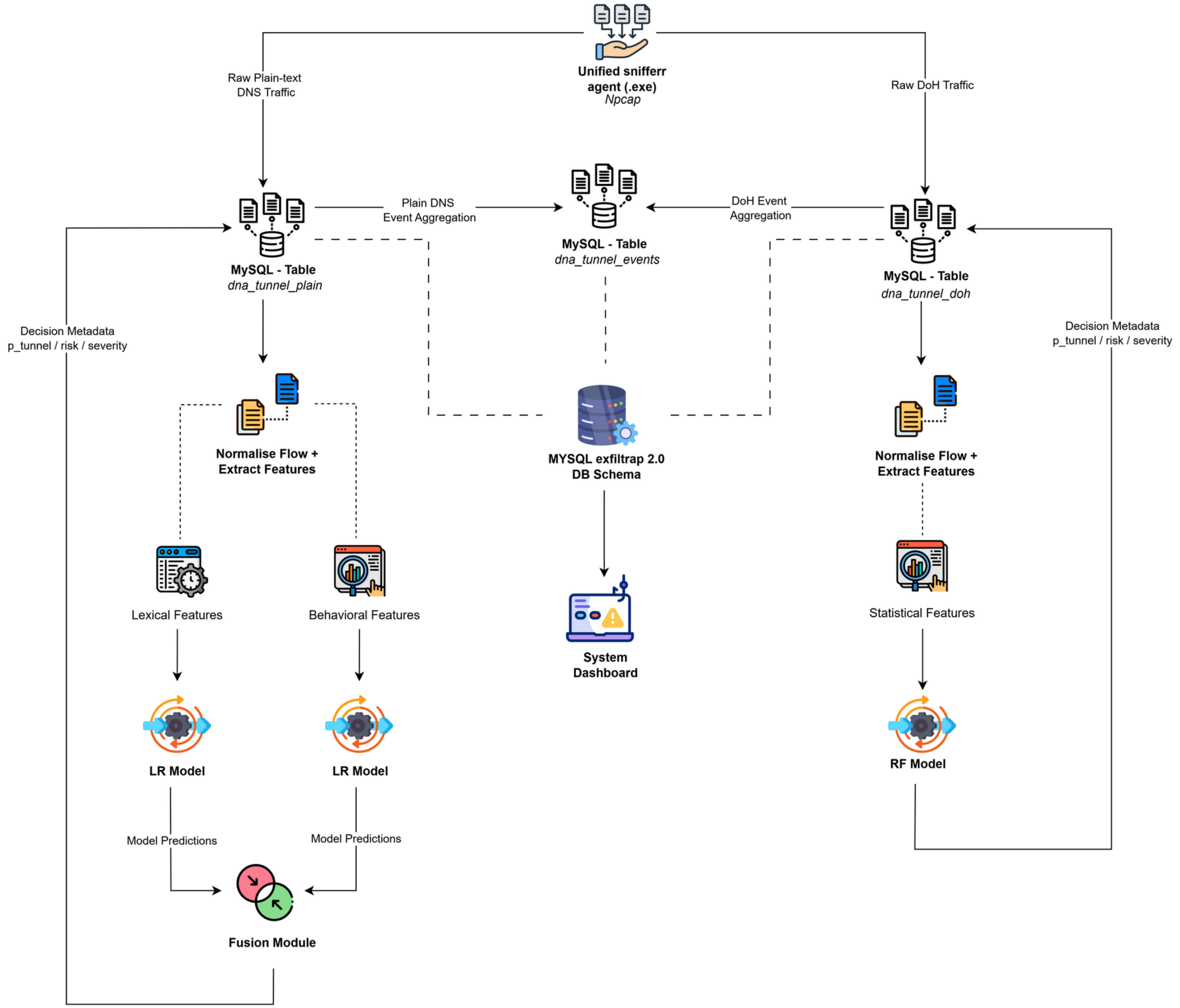
EXFILTRAP V2.0 DASHBOARD

[HTTP://APP.DCRYPTRA.COM](http://app.dcryptra.com)

Demo admin user credentials

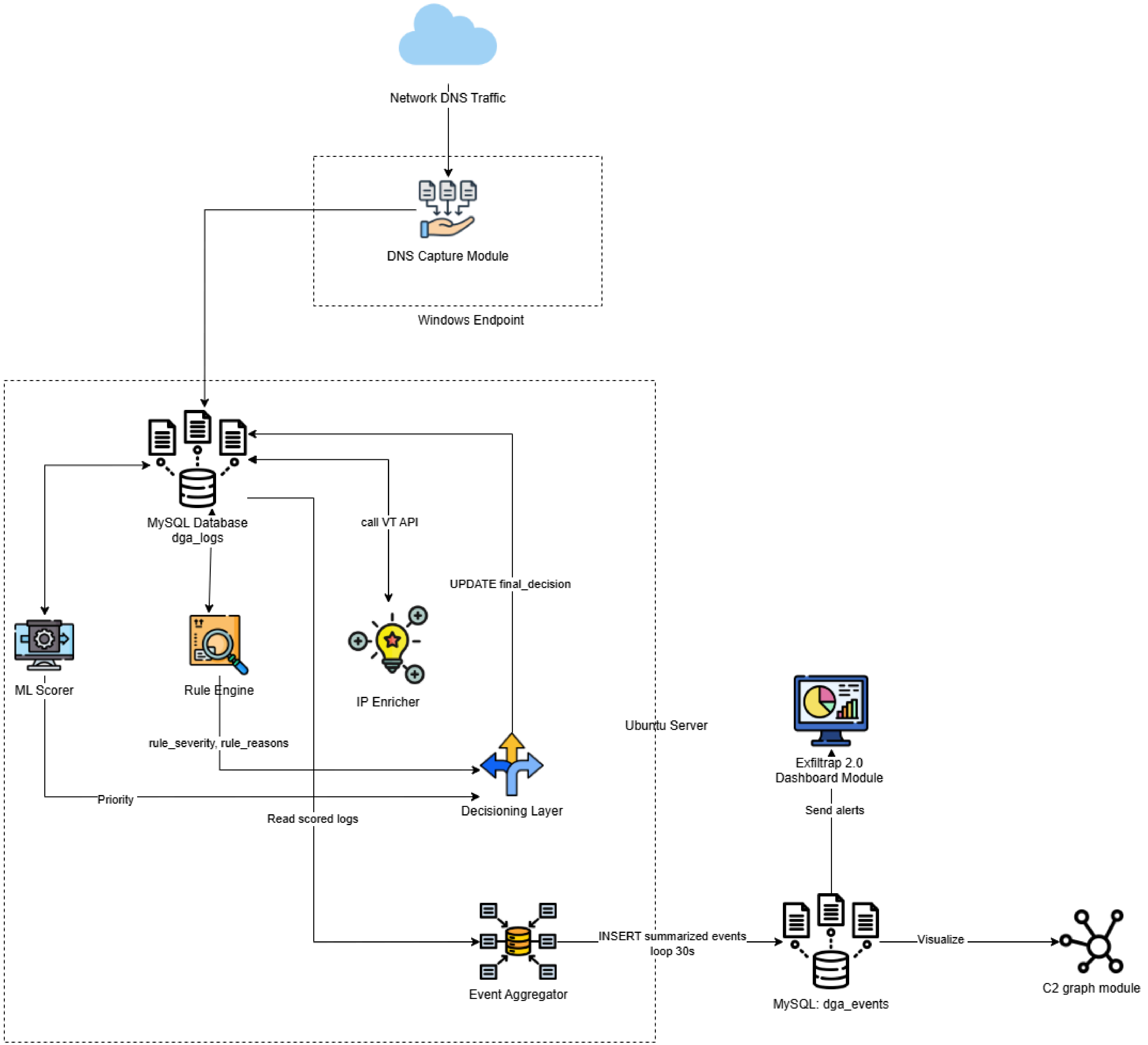
username: demouser-admin
password: demouser-admin@123

AI/ML DETECTION OF DNS TUNNELING & DATA EXFILTRATION

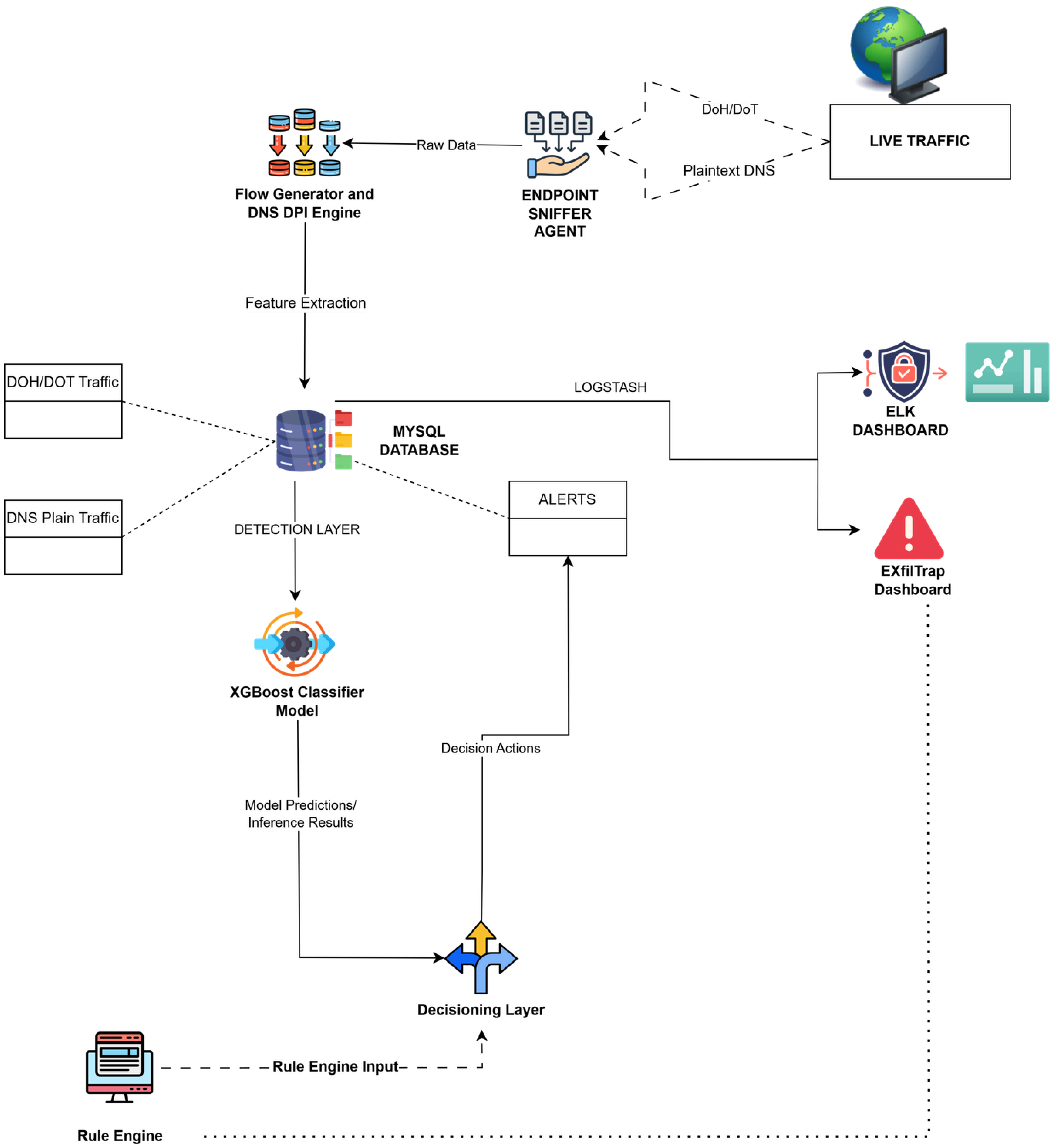


SLIIT
FACULTY OF COMPUTING

AI/ML DETECTION OF MALWARE C2 & DGA-BASED DNS



AI/ML DETECTION OF DNS ABUSE & INFRASTRUCTURE ATTACKS



SLIIT FACULTY OF COMPUTING

THREAT INTELLIGENCE & RESILIENCE MODULE